

VISHNU ASUTOSH DASU

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EDUCATION

- **Pennsylvania State University** *Aug 2024 - May 2028*
Doctor of Philosophy, Computer Science and Engineering
 - *Research Area:* Trustworthy AI and Software Security
 - *Advisor:* Prof. Gang (Gary) Tan
- **Pennsylvania State University** *Aug 2022 - May 2024*
Master of Science, Computer Science and Engineering
 - *Thesis:* “Mitigating Unfairness in Deep Learning”
 - *Advisor:* Prof. Gang (Gary) Tan
- **Manipal Institute of Technology (MIT), Manipal** *July 2016 - July 2020*
Bachelor of Technology, Computer Science and Engineering

SELECTED PUBLICATIONS

* - Equal Contribution/Alphabetical Order

- **Attention Pruning: Automated Fairness Repair of Language Models via Surrogate Simulated Annealing [pdf]**
48th International Conference on Software Engineering (ICSE), 2026
Vishnu Asutosh Dasu, Md Rafi ur Rashid, Vipul Gupta, Saeid Tizpaz-Niari, Gang Tan
- **Improving Noise Efficiency in Privacy-preserving Dataset Distillation [pdf]**
International Conference on Computer Vision (ICCV), 2025
Runkai Zheng, Vishnu Asutosh Dasu, Yinong Wang, Haohan Wang, Fernando De la Torre
- **Privacy-Preserving Data Deduplication for Enhancing Federated Learning of Language Models [pdf]**
Network and Distributed System Security (NDSS) Symposium, 2025
Aydin Abadi*, Vishnu Asutosh Dasu*, Sumanta Sarkar*
- **Chain-of-Thought Driven Adversarial Scenario Extrapolation for Robust Language Models [pdf]**
40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026
Md Rafi Ur Rashid, Vishnu Asutosh Dasu, Ye Wang, Gang Tan, Shagufta Mehnaz
- **NeuFair: Neural Network Fairness Repair with Dropout [pdf]**
33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2024
Vishnu Asutosh Dasu, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan
- **Gradient-Free Privacy Leakage in Federated Language Models through Selective Weight Tampering [pdf]**
Privacy Enhancing Technologies Symposium (PETS) 2026
Md Rafi Ur Rashid, Vishnu Asutosh Dasu, Kang Gu, Najrin Sultana, Shagufta Mehnaz
- **PROV-FL: Privacy-preserving Round Optimal Verifiable Federated Learning [pdf]**
15th ACM Workshop on Artificial Intelligence and Security (AISEC), ACM CCS, 2022
Vishnu Asutosh Dasu, Sumanta Sarkar, Kalikinkar Mandal
- **Side Channel Attack On Stream Ciphers: A Three-Step Approach To State/Key Recovery [pdf]**
IACR Conference on Cryptographic Hardware and Embedded Systems (CHES), 2022

Satyam Kumar, **Vishnu Asutosh Dasu**, Anubhab Baksi, Santanu Sarkar, Dirmanto Jap, Jakub Breier, Shivam Bhasin

- **Impact of Data Duplication on Deep Neural Network-Based Image Classifiers: Robust vs. Standard Models** [pdf]
IEEE Deep Learning Security and Privacy Workshop (DLSP), IEEE S&P, 2025
Alireza Aghabagherloo, Aydin Abadi, Sumanta Sarkar, **Vishnu Asutosh Dasu**, Bart Preneel
- **EvoquerBot: A multimedia chatbot leveraging synthetic data for cross-domain assistance** [pdf]
Alexa Prize TaskBot Challenge 2 Proceedings
Team EvoquerBOT, Penn State University
- **New Results on Machine Learning-Based Distinguishers** [pdf]
IEEE Access, 2023
Anubhab Baksi*, Jakub Breier*, **Vishnu Asutosh Dasu***, Xiaolu Hou*, Hyunji Kim*, Hwajeong Seo*
- **Three Input Exclusive-OR Gate Support For Boyar-Peralta’s Algorithm** [pdf]
22nd International Conference on Cryptology in India (Indocrypt), 2021
Anubhab Baksi, **Vishnu Asutosh Dasu**, Banashri Karmakar, Anupam Chattopadhyay, Takanori Isobe
- **LIGHTER-R: Optimized Reversible Circuit Implementation For SBoxes**
32nd IEEE International System-on-Chip Conference (SOCC), 2019
Vishnu Asutosh Dasu, Anubhab Baksi, Sumanta Sarkar, Anupam Chattopadhyay
- **[Re] GANSpace: Discovering Interpretable GAN Controls**
ReScience C, 2022
Vishnu Asutosh Dasu, Midhush Manohar T.K.

PREPRINTS/UNDER REVIEW

- **Heimdall: Formally Verified Automated Migration of Legacy eBPF Programs to Rust**
(Under Review) ACM Conference on Computer and Communications Security (CCS) 2026
Vishnu Asutosh Dasu, Monika Santra, Md Rafi Ur Rashid, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan
- **Robustness of Vision Language Models Against Split-Image Harmful Input Attacks**
(Under Review) ACM Conference on Computer and Communications Security (CCS) 2026 [pdf]
Md Rafi Ur Rashid, Md Sadik Hossain Shanto, **Vishnu Asutosh Dasu**, Shagufta Mehnaz
- **iResolveX: Multi-Layered Indirect Call Resolution via Static Reasoning and Learning-Augmented Refinement** [pdf]
(Under Review) Network and Distributed Systems Security Symposium (NDSS) 2027
Monika Santra, Bokai Zhang, Mark Lim, **Vishnu Asutosh Dasu**, Dongrui Zeng, Gang Tan
- **Finding Probabilistic Sub-invariants via Search Space Algorithms under Noisy Cost Functions**
(Under Review) Formal Methods in Computer-Aided Design (2026)
Ashish Kumar, **Vishnu Asutosh Dasu**, Saeid Tizpaz-Niari, Gang Tan

UNDER SUBMISSION

- **Half the Embeddings Do Nothing!**
(Under Submission) Neural Information Processing Systems (NeurIPS) 2026
Ashish Kumar, **Vishnu Asutosh Dasu**, Jialun Zhang, Ranit Debnath Akash, Saeid-Tizpaz Niari, Gang Tan

- **[Placeholder Title] Code Hallucinations in Large Language Models**
(Under Submission) *Neural Information Processing Systems (NeurIPS) 2026*
Vishnu Asutosh Dasu, Ashish Kundu, Ramana Kompella, Gang Tan
- **ALTO: Enforcing Information-Flow Policies for MCP Agents via Fine-Grained Context-Based Access Control**
In Progress
Vishnu Asutosh Dasu, Monika Santra, Ashish Kumar, Arslan Khan, Gang Tan

ACADEMIC AND WORK EXPERIENCE

- **Pennsylvania State University** *Aug. 2022 - Present*
Graduate Research/Teaching Assistant *Advisors: Gang Tan, Saeid Tizpaz-Niari, Shagufta Mehnaz*
– *Projects:* Data extraction attacks on federated LLMs, Trustworthy Code Generation LLMs, Fairness and Robustness of LLMs, Binary Analysis, Software Security
- **Tata Research Development and Design Centre (TRDDC)** *Sept 2020 - June 2022*
Researcher, Cybersecurity and Privacy *Advisors: Sumanta Sarkar*
– *Projects:* Privacy-preserving Federated Learning, Secure Multiparty Computation, Insider Threat Detection
- **Citrix R&D** *Jan 2020 - June 2020*
Software Engineer Intern, Citrix Analytics for Security (CAS)
– Full-stack Web Developer (Spring Boot, React.js)
- **Nanyang Technological University (NTU)** *Dec 2019 - Jan 2020*
Research Intern *Advisors: Anupam Chattopadhyay*
– *Projects:* Optimizing hardware implementations of block ciphers
- **Tata Consultancy Services (TCS) Research** *May 2019 - July 2019*
Research Intern, Cybersecurity and Privacy *Advisors: Chalamala Srinivasa Rao*
– *Projects:* Adversarial attacks and defenses on Convolution Neural Networks (CNNs)
- **Indian Statistical Institute** *May 2018 - July 2018*
Summer Scholar *Advisors: Dipti Prasad Mukherjee*
– *Project:* 3-D coordinate estimation of humans from 2-D live video feed

SKILLS

- **Languages:** Python, C/C++, Java, Rust, HTML/CSS, Go, JavaScript, SQL
- **AI/ML Expertise:** Machine Learning, Deep Learning, Large Language Models (LLMs), Trustworthy AI (Fairness, Privacy, Robustness), NLP, Computer Vision
- **Tools:** PyTorch, TensorFlow, Huggingface, GMP, OpenSSL, L^AT_EX, KLEE, LLVM, Angr, Git
- **Areas of Expertise:** Software & Hardware Security, Privacy, Applied Cryptography

SERVICE

- **Reviewer:** AAAI 2026, IEEE TDSC (2026), IEEE TIFS (2025), IEEE Access (2025)
- **Artifact Evaluation Committee:** ACM CCS 2026, ACM CCS 2025, USENIX Security 2026
- **External Reviewer:** NDSS 2025, OOPSLA 2025, IEEE S&P 2025, IEEE S&P 2026
- **Organizer:** Penn State Security Reading Group (2024 - Present)

- **Judge:** Penn State Undergraduate Exhibition 2025

AWARDS AND ACHIEVEMENTS

- **Student Travel Grant:** Received the ACM/SIGSOFT CAPS travel grant to attend ICSE 2026.
- **Distinguished Reviewer Award:** Received the distinguished reviewer award for artifact evaluation in ACM CCS 2025.
- **Student Travel Grant:** Received the travel grant to attend ICCV 2025.
- **Internet Society Fellowship:** Received the Internet Society Fellowship to attend NDSS 2025.
- **TCS Citation Award** ($3 \times$ recipient): Received the award for outstanding contribution to TCS.
- **Scholarship:** Received a scholarship to attend the *Winter School on Responsible AI* in Israel.
- **Best Project Award:** Received the Best Project Award during the *Fifth Summer School on Computer Vision, Graphics and Image Processing*, Indian Statistical Institute (ISI) Kolkata.
- **IGVC:** Placed 2nd in the Interoperability Profiles Challenge and 9th overall at *Intelligent Ground Vehicle Competition (IGVC)* 2018. Second-best among all teams from India.
- **DAGsHub Award:** Received a \$500 award from *DAGsHub* for completing the *ML Reproducibility Challenge Spring 2021*.

PROFESSIONAL MEMBERSHIPS

- Student Member of the ACM
- Student Member of IEEE

REFERENCES

- Prof. Gang Tan (Advisor): gtan@psu.edu
- Prof. Saeid Tizpaz-Niari: saeid@uic.edu
- Prof. Sumanta Sarkar: sumanta.sarkar@essex.ac.uk